and operating system trust stores, ensuring seamless and trusted connectivity.

Am I Impacted?

This change could impact **B2B Connectivity**:

The sender systems must trust both the old root certificate (*DigiCert Global Root G2*) and the new root certificate (*DigiCert TLS RSA4096 Root G5*). Without this, the inbound communication to SAP Business Network Freight Collaboration and SAP Business Network Global Track and Trace will be impacted.

For information about a possible impact on the end user UI, see Browser Support.

Resolution for B2B Connectivity to BN4L:

Case 1: When sender system is S/4HANA (On-Premise & Private Cloud)

1. Download [DigiCert TLS RSA4096 Root G5].
2. Upload this Root Certificate to all relevant SSL Client PSEs*.
3. For detailed steps, see SAP KBA 3238733 - How to Import a Certificate Into a STRUST PSE's Certificate List In a NetWeaver AS ABAP.
4. [DigiCert Global Root G2] should remain as is.
5. ICM restart should be necessary post change in STRUST, only if you are running old releases of Netweaver 700/701 and 64x. See SAP KBA 2746647 - SSSLERR_PEER_CERT_UNTRUSTED after importing the target certificate. Since Netweaver AS ABAP 710 and AS ABAP 702, ICM restart is no longer necessary.

Case 2: When sender system is SAP NetWeaver PI (On-Premise & Private Cloud)

1. Download [DigiCert TLS RSA4096 Root G5].
2. Import this Root Certificate to TrustedCAs view of NetWeaver Java Key Storage.
3. For detailed steps, see SAP KBA 2056672 - How to import server certificates in PI system.

Case 3: When WebDispatcher is placed in front of Cloud Integration endpoint

1. Download [DigiCert TLS RSA4096 Root G5].
2. Import this Root Certificate to Trusted Certificates through Web Administration Interface of SAP WebDispatcher.
3. See SAP Help - PSE Management in Web Administration Interface of SAP Web Dispatcher, section [Trusted Certificates].

Case 4: When sender system is a non-SAP service/software

1. Consult your vendor for confirmation that it should trust [DigiCert TLS RSA4096 Root G5] for outbound TLS communication.
2. Particularly, in case of a customer written Java program, JDK version should be higher than the following:
    - SAP JVM 8.1.097
    - SapMachine 11.0.22
    - SapMachine 17.0.10
    - SapMachine 21.0.2

- Oracle JDK 8u401
- Oracle JDK 17.0.10
- OpenJDK 17.0.10
- OpenJDK 21.0.2

For more information, see the following SAP Notes:

- 3688060
- 3587968
- 3566727
- 3472211

Connecting Directly Between SAP Business Network for Logistics and Your SAP Backend System

Perform the following steps:

1. Configuring Inbound to SAP Business Network for Logistics
2. Configuring in SAP Backend Systems via SOA Manager
3. Activate the connection in SAP Business Network for Logistics. (Refer to the document Activating the Connection.)

Generally, you use a service user or RFC user to communicate with the SAP TM or SAP S/4HANA system. This user processes the incoming B2B messages and must have the following authorization objects so that they can create or update the correct documents in the external system:

- In SAP TM, the user must have the authorization objects T_ADMIN and T_TM_ALL.
- In SAP S/4HANA, the user must have the authorization objects T_ADMIN, T_TOR_EXE and S_SERVICE.

For more details, refer to the following SAP Note: 2890717.

You can add these authorization objects to the role assigned to the service user. For more details, refer to the following SAP Notes: 440231, 1696484.

For more information about direct connections to SAP Business Network for Logistics from:

- SAP S/4HANA, refer to the following blog: https://blogs.sap.com/2021/03/16/setting-up-connection-between-sap-s-4hana-and-sap-logistics-business-network-global-trace-and-trace-option/.
- SAP ERP, refer to the following blog: https://blogs.sap.com/2021/07/07/how-to-replicate-location-from-sap-erp-to-sap-logistics-business-network/?update=publish.

Certificates

Before communication from and to your SAP backend system can take place, authentication with public and private keys, or certificates, must take place.

Communication from and to your SAP backend system is based on B2B messages using SOAP and REST protocols. For authentication at SAP Business Network for Logistics, client certificates are used. Therefore, these certificates must be requested as a prerequisite. Several possible ways in which such key pairs (public and private key, sometimes also just called certificates) can be generated are listed below. A `*.p12` file will be generated as a result. You may use any of the following options:

- Private and public key creation with the Identity Authentication service
- SAP Cloud Integration: keystore
- Trust manager – STRUST
- Private and public key creation with a third-party tool

Creating a Private and Public Key with the Identity Authentication Service

Before communication from and to your SAP backend system can take place, authentication with public and private keys, or certificates, must take place. The identity authentication service is one option for doing this.

Perform the following steps:

1. Obtain access to the Identity Authentication service.

   There are three possible ways in which you can achieve this:

   - You have already purchased an Identity Authentication service tenant. You can purchase such a service tenant here: < https://www.sapstore.com/solutions/40132/SAP-Cloud-Platform-Identity-Authentication>.
   - While subscribing to a productive license for SAP Business Network Global Track and Trace, you have been provisioned with an Identity Authentication service tenant and details tenant, and a URL is sent to the S-User used for the license subscription.
   - If you have subscribed for a test license for SAP Business Network Global Track and Trace, and you have not purchased an Identity Authentication service tenant, you may request a key pair from SAP by raising an incident to the component `SBN-LBN-INT`. (In this case, you can skip the steps in this section.)

   When using the Identity Authentication service, the certificates are signed by SAP Passport CA.

2. Follow the steps below to generate a `*.p12` file from your Identity Authentication service tenant.

   Perform the following actions to generate a key pair. The following process is only for an SAP Business Network for Logistics productive license.

   a. Access the tenant's administration console for the Identity Authentication service by using the console's URL.

      Note the following points:

      - The URL has the pattern https://<tenant ID>.accounts.ondemand.com/admin.
      - The tenant ID is automatically generated by the system. The first administrator who created the tenant receives an activation email with a URL. This URL contains the tenant ID.

   b. Under **Applications and Resources**, choose **Applications**, click the pencil icon for **Add Application**, and assign the new application the name `CertificateGeneration`, for example. Then search for **CertificateGeneration** and click on **Certificate for API Authentication**.

   c. Enter the **Common Name**, **Password**, and **Confirmed Password** and click on **Generate**. The browser downloads the certificate to your local system.

The result is that you will have a signed certificate – a `*.p12` file.

Further information is available here: API Authentication

SAP Cloud Integration: Keystore

Before communication from and to your SAP backend system can take place, authentication with public and private keys, or certificates, must take place. The SAP Cloud Integration keystore is one option for doing this.

If you have access to an SAP Cloud Integration tenant, you can use the available keystore functionality. Further information on generating the certificate is available here: Creating a Key Pair/SSH Key Pair.

Trust Manager – STRUST

Before communication from and to your SAP backend system can take place, authentication with public and private keys, or certificates, must take place. Trust manager – STRUST is one option for doing this.

Follow the steps described in Configuring SAP NetWeaver AS for ABAP to Support TLS to configure STRUST certificates. Here, you will also find the information on Personal Security Environment (PSE) creation and certificates. Further information is available here: Configuring the Web Service Runtime.

For additional information on this topic, check out this SAP blog on How to order an SSL certificate.

Creating a Private and Public Key with a Third-Party Tool

Before communication from and to your SAP backend system can take place, authentication with public and private keys, or certificates, must take place. Third-party tools are available for doing this.

Create a new `*.p12` file (private key and certificate), such as OpenSSL, and get it signed by a supported certificate authority (CA).

Certificates created using the Identity Authentication service are already signed by SAP Passport CA. Any other certificates which are not yet signed by a CA need to be signed by one of the SAP Business Network for Logistics-supported CAs.

Third-party certification authorities

- AddTrust External CA Root
- Baltimore CyberTrust Root
- Certum CA
- DigiCert High Assurance EV Root CA
- DigiCert Global Root CA
- GeoTrust Global CA
- GeoTrust Primary Certification Authority - G3
- GlobalSign
- GlobalSign Root CA
- Go Daddy Class 2 Certification Authority

- Go Daddy Root Certificate Authority - G2
- Entrust Root Certification Authority
- Entrust.net Certification Authority (2048)
- Entrust Root Certification Authority - G2
- Equifax Secure Certificate Authority
- QuoVadis Root CA 2
- TC TrustCenter Class 2 CA II
- SwissSign Gold CA - G2
- SwissSign Platinum CA - G2
- SwissSign Silver CA - G2
- Thawte Primary Root CA
- Thawte Primary Root CA - G3
- VeriSign Class 1 Public Primary Certification Authority - G3
- VeriSign Class 3 Public Primary Certification Authority
- VeriSign Class 3 Public Primary Certification Authority - G5
- VeriSign Universal Root Certification Authority

For **additional information** on this topic, check out this SAP blog on How to order an SSL certificate.

Configuring Inbound to SAP Business Network for Logistics

Extracting a Client Certificate from a *.p12 File

Use the `*.p12` file and KeyStore Explorer (https://keystore-explorer.org/) to extract the public certificate as follows:

1. Open the `*.p12` file and export the client certificate, which should have your company's common name (CN). Make sure that your company's certificate is extracted and not one of the CA certificates.
2. Export the certificate in `X.509` format and save as a `*.cer` file.

If you have created a new PSE in STRUST, the certificate can also be downloaded from the corresponding PSE within STRUST.

For more information on exporting and importing a PSE from and to STRUST, see SAP Note 1473710.

Uploading the Certificate to SAP Business Network for Logistics via System Connection App

Perform the following steps:

1. Creating a Connection.

2. Upload the certificate to SAP Business Network for Logistics. You can refer to the application help here: Authenticating Inbound Messages to SAP Business Network for Logistics.

Configuring in SAP Process Integration or SAP Cloud Integration to Maintain Certificate and Links

This is a prerequisite for SAP Process Integration:

Configuring Business Systems with Integration Engine	https://help.sap.com/viewer/6dbe2ddcde6a4087858c533f8032445b/7.5.16/en-US/48a9bc5d7e28674be10000000a421937.html
Configuring the Use of SSL on the AS Java (connectivity via SAP Process Integration)	https://help.sap.com/viewer/a42446bded624585958a36a71903a4a7/7.5.5/en-US/4a015cc68d863132e10000000a421937.html

Uploading Certificates

Depending on which method you choose, you need to upload the certificate to SAP Process Integration, SAP Cloud Integration, SAP Transportation Management (SAP TM) standalone, or Transportation Management embedded in SAP S/4HANA.

Uploading Certificates to SAP Process Integration

Perform the following steps:

1. From the `*.p7b` file, extract the root, intermediate and leaf certificates (`*.cer`) using a third-party tool like the KeyStore Explorer tool (https://keystore-explorer.org). Afterwards upload them to the Trusted CAs keystore of SAP Process Integration.

2. Upload the `*.p12` file (key pair) to the Trusted CA of SAP Process Integration.

3. Download the root and intermediate runtime SSL certificates from the host URL depending on your SAP Business Network for Logistics tenant's data center, and then upload them to the Trusted CA keystore of SAP Process Integration:

Host URLs by Data Center

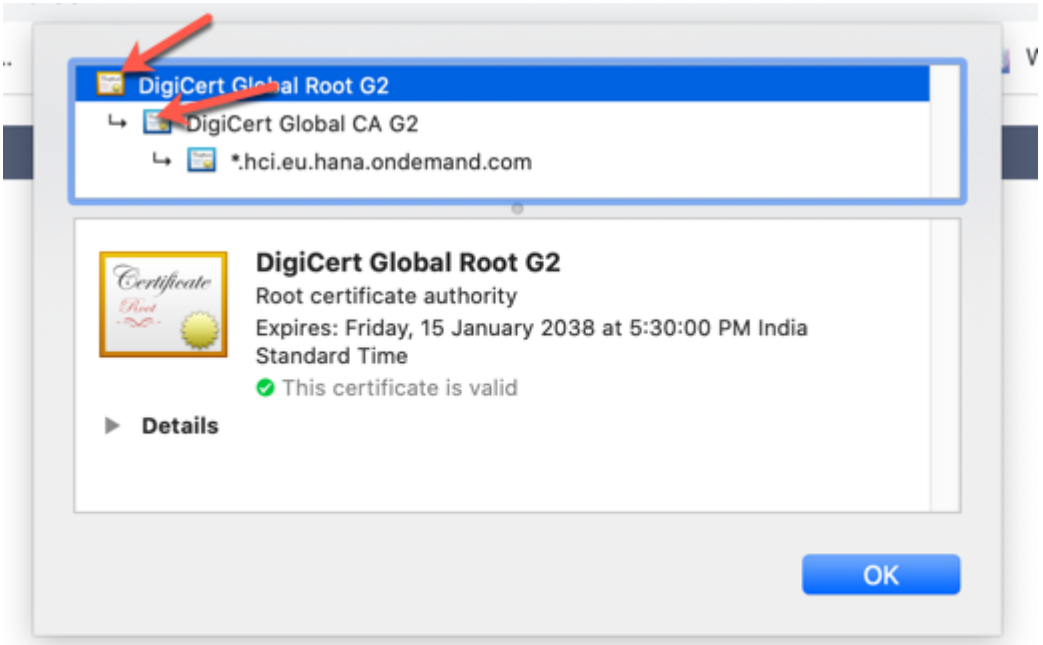
Data Center	Integration Platform Host URL of SAP Business Network for Logistics
EU	`https://api.logistics.eu.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`
US	`https://api.logistics.us.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`
KSA	`https://api.logistics.sa30.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`

Note

You only need to upload the certificate if the serial number is different to the certificate you uploaded in step one.

4. To view the site information, click on the padlock icon at the start of the URL.

5. In the popup indicating that the connection is secure, select **Certificate**.
6. Click on the root certificate (such as **DigiCert Global Root G5**), and download the certificate.
7. Click on the intermediate certificate (such as **DigiCert Global CA G5**), and download the certificate.
8. Upload the certificates to the Trusted CAs keystore of SAP Process Integration.



For **additional information** on this topic, check out this SAP blog on How to load keys and certificates in SAP PI 7.3, SAP PO 7.3 EHP1 SAP NetWeaver Administrator's key storage.

Uploading Certificates to SAP Cloud Integration

Perform the following steps:

1. Upload the `*.p12` file (key pair) to the SAP Cloud Integration tenant keystore.
2. Extract the root and intermediate runtime SSL certificate of the host URL depending on your SAP Business Network for Logistics tenant's data center, then upload to the SAP Cloud Integration tenant keystore.

Host URLs by Data Center

Data Center	Integration Platform Host URL of SAP Business Network for Logistics
EU	`https://api.logistics.eu.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`
US	`https://api.logistics.us.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`
KSA	`https://api.logistics.sa30.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`

ℹ Note

You only need to upload the certificate if the serial number is different to the certificate you uploaded in step one.

3. Extract all the certificates from the `*.p7b` file using an external application such as KeyStore Explorer.
4. Upload the certificates in the integration flow (iFlow) that you have configured to receive messages from SAP Business Network for Logistics.

Uploading Certificates to Your SAP Backend System

For information about uploading certificates to SAP S/4HANA or SAP TM, see the *Setting up the connection* section in the following blog: https://blogs.sap.com/2021/03/16/setting-up-connection-between-sap-s-4hana-and-sap-logistics-business-network-global-trace-and-trace-option/.

Maintaining an LBN Endpoint

Depending on your middleware – either SAP Process Integration or SAP Cloud Integration – you have to follow one of the following related sections:

- Maintaining an LBN Endpoint in SAP Process Integration
- Maintaining an LBN Endpoint in SAP Cloud Integration

Maintaining an LBN Endpoint in SAP Process Integration

In the target URL of your integration system, maintain the URL based on your SAP Business Network for Logistics tenant's data center to forward the messages to the SAP Cloud Integration system.

Target URLs by Data Center

Data Center	Integration Platform Target URL of SAP Business Network for Logistics
EU	`https://api.logistics.eu.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`
US	`https://api.logistics.us.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`
KSA	`https://api.logistics.sa30.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`

You must configure the following interfaces in your integration system:

Outbound interfaces

- LocationBulkReplicationRequest_Out

The sample configuration is just a high-level overview on one message type without message mapping on SAP Process Integration. Several other options to configure the message routing are outlined below.

Receiver Determination:

- The interface and interface namespace are required for the message type to be forwarded to SAP Business Network for Logistics.
- You must specify the receiver communication component representing SAP Business Network for Logistics (with or without a routing condition).

Receiver Agreement:

- For the determined receiver and interface, specify the technical communication channel which is connected to SAP Business Network for Logistics.

Receiver Communication Channel:

- Select the SOAP adapter type.
- The receiver communication channel needs to have the following target URL based on your SAP Business Network for Logistics tenant's data center:

Target URLs by Data Center

Data Center	Integration Platform Target URL of SAP Business Network for Logistics
EU	`https://api.logistics.eu.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`
US	`https://api.logistics.us.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`

- The keystore entry and keystore view must reference the certification key which should be used for authentication at SAP Business Network for Logistics.
- If a connection to SAP Business Network for Logistics can only be established via a proxy, specify your company's proxy settings.

Configuring in SAP Backend Systems via SOA Manager

This is a prerequisite for Web Service Runtime:

Configuring SAP NetWeaver AS for ABAP to Support TLS	https://help.sap.com/viewer/e73bba71770e4c0ca5fb2a3c17e8e229/7.5.16/en-US/4923501ebf5a1902e10000000a42189c.html
Configuring the Web Service Runtime	https://help.sap.com/viewer/684cffda9cbc4187ad7dad790b03b983/7.5.16/en-US/47ab7d3b49f84848e10000000a421138.html?q=Configuring%20the%20Web%20Service%20Runtime

Setting Up Trust

Web Service Runtime (SAP ABAP)

ABAP Web Services

SAP NetWeaver Application Server for ABAP provides a standardized architecture and a set of tools for creating Web services and related objects. Existing BAPIs or remote-enabled function modules can be used for setting up Web services, or you can develop new Web services in the Object Navigator of SAP NetWeaver AS for ABAP. There are predefined settings for securing Web services or the transports being used. ABAP Web services can be used for communication between SAP systems and between SAP and non-SAP systems. When using the Web Service runtime, this is a direct connection from the SAP backend system to SAP Business Network for Logistics with no middleware involved.

Configuring HTTPS at Transport Level with X.509 Certificate Authentication

For information on how to configure HTTPS at transport level, see the following document on SAP Help Portal: Configuring HTTPS at Transport Level with X.509 Certificate Authentication.

Using a New PSE in SAP Backend System

Create a new PSE in STRUST which contains the relevant private key and certificates required for communicating with SAP Business Network for Logistics. To do so, perform the following steps:

1. Go to the transaction STRUST.
2. In the menu bar, under the option **Environment**, choose **SSL Client Identities**.
3. Create a new entry for SAP Business Network for Logistics.
4. Save your settings.

Trust Manager

For information on how to establish trust relationships with the trust manager, see the following document on SAP Help Portal: Trust Manager.

You already have a signed key pair (`*.p12` file) which you can import, or you create a new PSE.

To import the key pair, perform the following actions:

1. Go to the transaction STRUST.
2. Choose **PSE** from the menu options.
3. Choose **Import**.

Once you have imported the key pair, carry out the following steps:

1. Go to the menu option **PSE**.
2. Save as **SSL Client**.
3. Select the newly created folder (as defined in above step).

   As a result, you have now imported a separate certificate, which is later used for authentication in SAP Business Network for Logistics.

   For further information, refer also to SAP Knowledge Base Article (KBA) 1850809: Reorder a certificate chain using Netweaver Administrator.
4. Download the runtime SSL certificate from the following URL depending on your SAP Business Network for Logistics tenant's data center:

Host URLs by Data Center

Data Center	Integration Platform Host URL of SAP Business Network for Logistics
EU	`https://api.logistics.eu.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`
US	`https://api.logistics.us.business-network.cloud.sap/cxf/lbn/b2b/soap/v1`